



Advanced Algorithms

COMP4121

Aleks Ignjatović

School of Computer Science and Engineering
University of New South Wales

Introduction

What are we going to do in this class?

- We will start with an introduction to randomised algorithms.

What are we going to do in this class?

- We will start with an introduction to randomised algorithms.
- We will also cover a sample of algorithms important for the present day practice, such as the Google's PageRank algorithm, algorithms for the recommender systems and, most importantly, a few algorithms used in Data Science.

What are we going to do in this class?

- We will start with an introduction to randomised algorithms.
- We will also cover a sample of algorithms important for the present day practice, such as the Google's PageRank algorithm, algorithms for the recommender systems and, most importantly, a few algorithms used in Data Science.
- We will cover parts of three books: *Algorithm Design* by Kleinberg and Tardosh, *Networked Life* by Mung Chiang (both available at the UNSW book store) and *Foundations of Data Science* by Blum, Hopcroft and Kannan whose preprint is available online for free at <https://www.cs.cornell.edu/jeh/book.pdf>.

What are we going to do in this class?

- We will start with an introduction to randomised algorithms.
- We will also cover a sample of algorithms important for the present day practice, such as the Google's PageRank algorithm, algorithms for the recommender systems and, most importantly, a few algorithms used in Data Science.
- We will cover parts of three books: *Algorithm Design* by Kleinberg and Tardosh, *Networked Life* by Mung Chiang (both available at the UNSW book store) and *Foundations of Data Science* by Blum, Hopcroft and Kannan whose preprint is available online for free at <https://www.cs.cornell.edu/jeh/book.pdf>.
- While we will focus on **practical** algorithms, these algorithms will be quite sophisticated and, just as any other piece of sophisticated engineering, involve some basic mathematics and some basic statistics.

What are we going to do in this class?

- We will start with an introduction to randomised algorithms.
- We will also cover a sample of algorithms important for the present day practice, such as the Google's PageRank algorithm, algorithms for the recommender systems and, most importantly, a few algorithms used in Data Science.
- We will cover parts of three books: *Algorithm Design* by Kleinberg and Tardosh, *Networked Life* by Mung Chiang (both available at the UNSW book store) and *Foundations of Data Science* by Blum, Hopcroft and Kannan whose preprint is available online for free at <https://www.cs.cornell.edu/jeh/book.pdf>.
- While we will focus on **practical** algorithms, these algorithms will be quite sophisticated and, just as any other piece of sophisticated engineering, involve some basic mathematics and some basic statistics.
- Since probability can be a rather tricky matter, we start with some puzzles!

Problem: (Monty Hall Problem) A prize is placed behind one of three closed doors. You are allowed to choose one door. After you make your choice, the Quiz Host Monty Hall opens one of the remaining two doors and shows you that the prize is not behind that door (Monty knows where the prize is and always chooses the empty door). You are now given the following choice: you can either stick with the door you have chosen or you can switch to the other door which is still closed. What should you do?

Problem: (Three Cards Problem) There are three cards: one is red on both sides, one is blue on both sides and one is red on one side and blue on the other. You pick a card at random and then choose the side of the card also at random and look at it and it turns out that it is red. What is the probability that the other side is also red?

Problem: (Two Pubs) After work you go to the nearest train station. There is only one train stopping there, going East-West. At the Eastern Platform the train goes to the East; at the last stop there is the Eastern City Pub; at that stop the train turns around and goes back to the West. At the Western platform the train goes to the West, at the last stop is the Western City Pub; at that stop the train also turns around and goes back East. The trains go at perfect 5 minute intervals appart. You always choose the first train to arrive and go to the corresponding Pub. After a year you noticed that you were 4 times as often at the Eastern Pub than at the Western Pub. How is that possible?

Problem: (Unfair Coin) Assume that you have a biased coin, but you do not know the probability of getting a head. How could you get the equivalent of a fair coin with several tosses of the unfair coin?

Problem: (Money in Boxes) Someone shows you two boxes and he tells you that one of these boxes contains two times as much as the other one, but he does not tell you which one. He lets you choose one of these boxes, and opens it. It turns out to have \$10. Now he gives you the opportunity to change your mind and take only the money from the other box.

You reason as follows: *“The second box contains either half of the money of the first box or twice as much as the first box. Thus, it either contains \$20 with chance one half, or it contains only \$5 with chance also one half. Thus the expected value is $1/2 * \$20 + 1/2 * \$5 = \$12.5$. Consequently, since the expected value is larger than the value of the first box, I should switch.”*

Question: Is your reasoning correct?

To understand where the confusion in the previous example is coming from, let us consider a similar but different problem.

Problem: (Money in Boxes, a different version) Someone tells you that he will randomly chose a pair of numbers $x = 2^k$ and $y = 2^{k+1}$ with equal probability $1/9$ for all k such that $1 \leq k \leq 9$. Then he will place x dollars in one box and y dollars in another identical box. You get to chose one box and open it. After you see how much money is in the box you can decide to either keep that money or to take the money from the other, unopened box. What should be your strategy, should you switch the boxes?